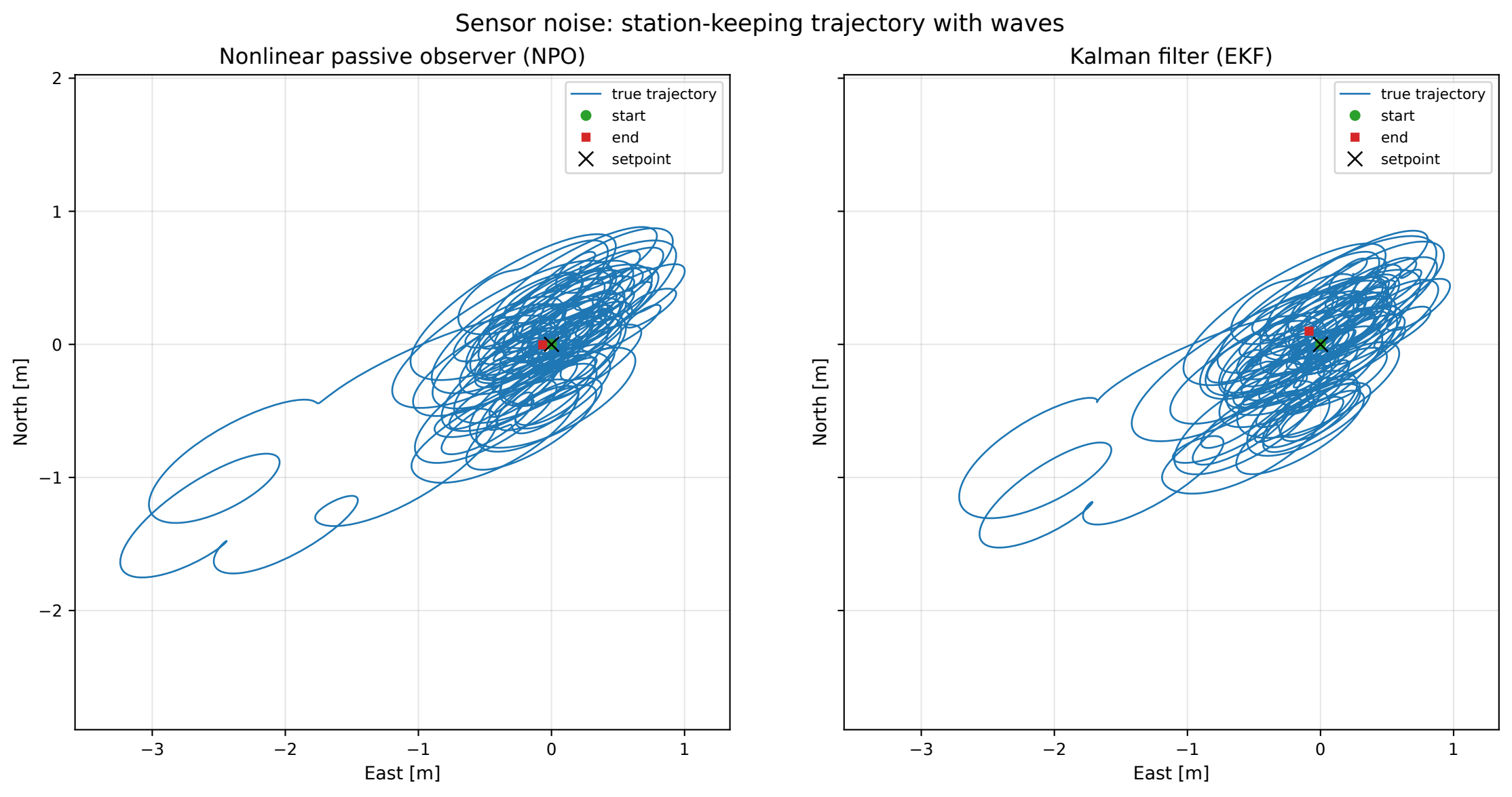

Sensor noise: station-keeping trajectory with waves
Nonlinear passive observer (NPO)
Kalman filter (EKF)
true trajectory
start
end
setpoint
North [m]
East [m]
2
1
0
−1
−2
−3
−2
−1
0
1